Module 4 Individual Assessment (Network Traffic Analysis)

Incident Forensic Investigation Report & Technical Assessment

PART 1: Executive Incident Summary

- **Prepared By: Mercy Itunuoluwa Olalere**
- **Incident Title:** Enterprise SMB Ransomware-in-Place Campaign (Project Enterprise 18.pcap)
- **Target Server:** FS01 (10.18.0.20)
- **Compromised Workstation:** 10.18.0.107
- **Incident Window:** October 7, 2018 ($t = 24,059.01s$ to $25,368.80s$)

Executive Overview

An investigation into Project Enterprise 18.pcap revealed a coordinated, automated ransomware attack targeting the private internal subnet (10.18.0.0/24). Workstation 10.18.0.107 generated 462 MB of traffic across 397,284 packets interacting heavily with the primary file server FS01 (10.18.0.20, which moved 487 MB across 417,116 packets).

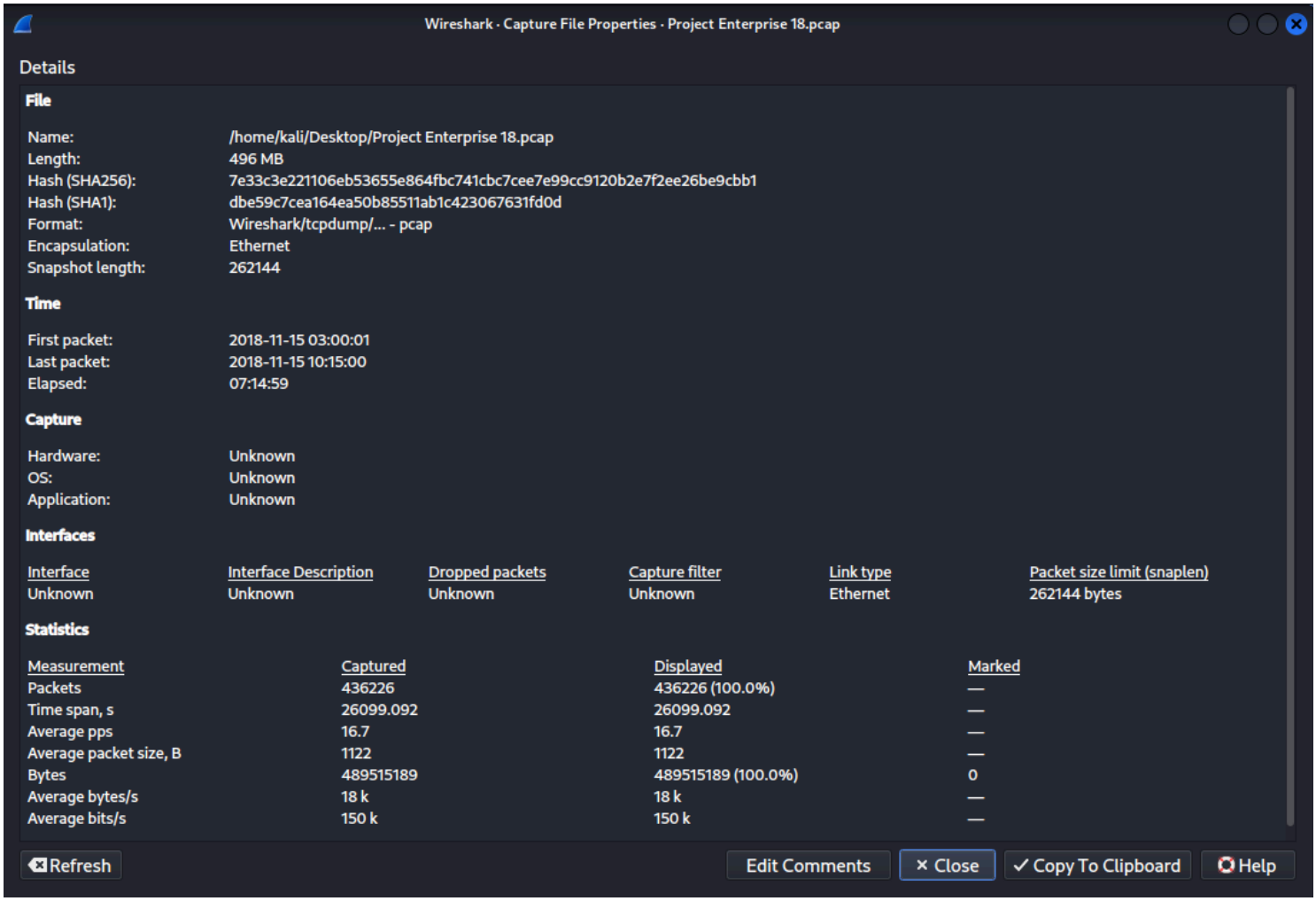
Following iterative NTLM authentication probes, the attacker established a valid session, mounted core departmental shares (\\FS01\Finance, \\FS01\HR, \\FS01\Projects, \\FS01\Legal, \\FS01\Departments), and executed over 17,000 rapid file operations.

Rather than exfiltrating data externally, the threat actor performed **client-side encryption-in-place**, utilizing temporary staging files (~RF*.tmp) to overwrite files with high-entropy ciphertext before appending the custom .arc2018 extension. All traffic remained strictly internal; background noise (such as RFC 5737 test range telemetry and external domains resolving to 198.51.100.x) was successfully filtered out as non-causal noise.

PART 2: Technical Report (Filters, Packet Evidence, Timeline & Screenshots)

1. Network Architecture & Scope (Tasks 2–4)

- **Internal Subnet:** 10.18.0.0/24 (Private RFC 1918 space).
- **Documentation/Test Ranges:** Observed external IPs (198.51.100.x, 203.0.113.x) confirmed as RFC 5737 test ranges.
- **Traffic Volume Evaluation:** Volume alone (462 MB) only proves data movement, requiring protocol-level SMB inspection to prove destructive encryption loops rather than routine backups.



2. DNS & HTTP Noise Reduction (Tasks 5–8)

- **Anomalous DNS Resolves:** Domains such as sowcar.com, z9ls.com, baocangwh.cn, gwjyhs.com, and w2wz.cn resolved to 198.51.100.x test IPs at wide, periodic intervals.
- **Causality Conclusion:** These domains were rejected as red herrings / background telemetry because their timestamps were decoupled from the dense SMB attack window. Standard software update queries (windowsupdate.com, microsoft.com) were likewise categorized as benign background noise.

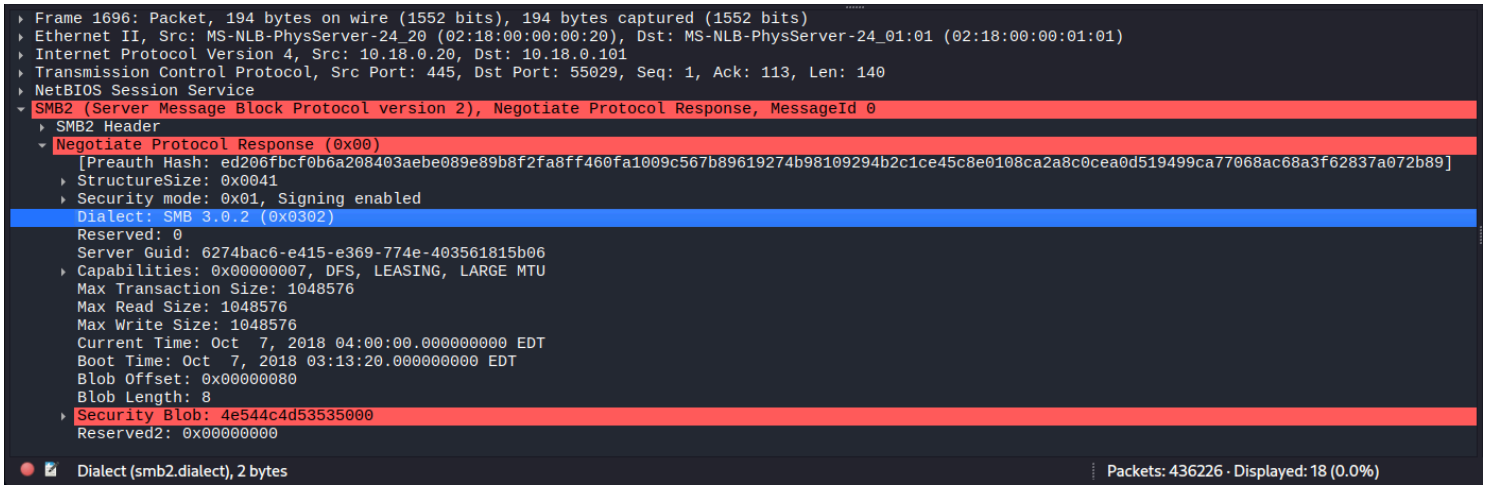
3. Core SMB Investigation & Evidence Placement (Tasks 9–18)

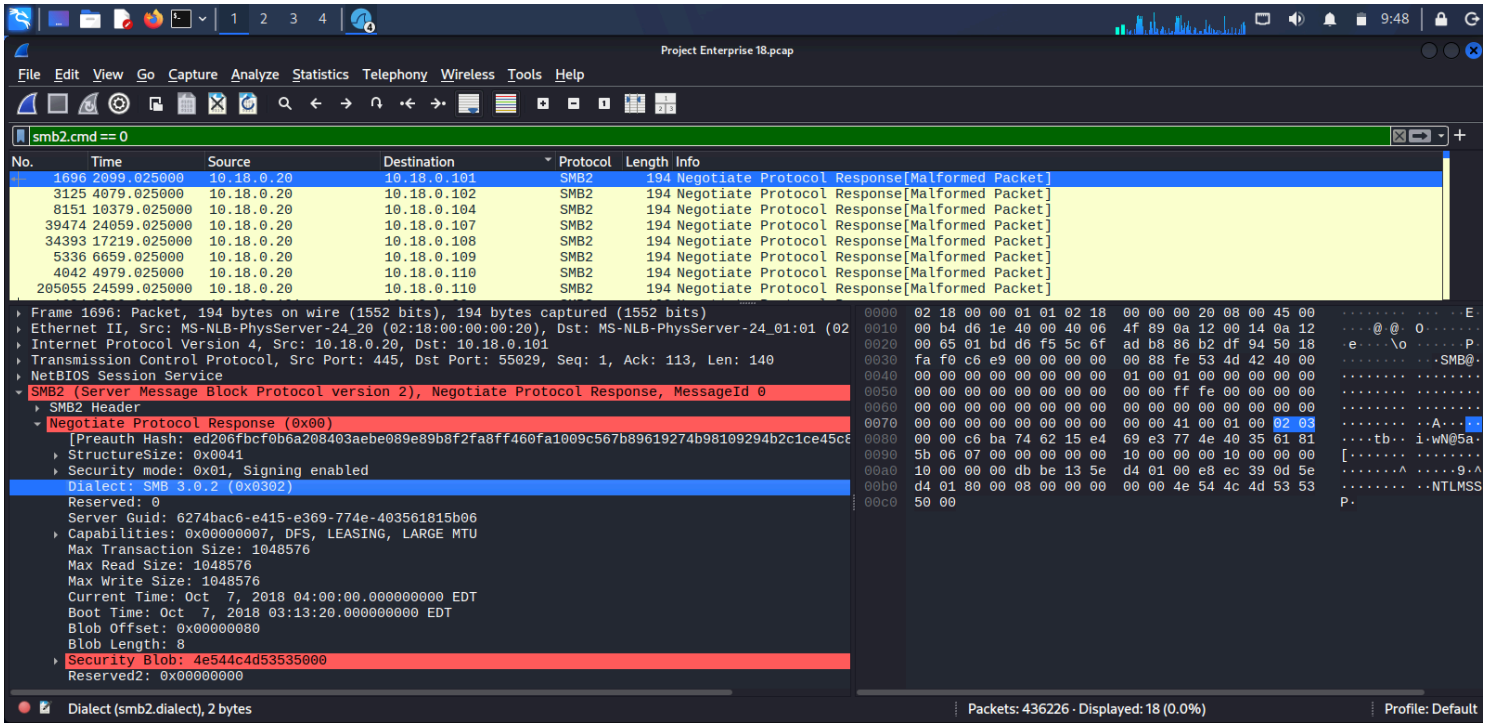
Stream 1404 Authentication Progression



- **Context:** This visually support **Task 11 (Authentication Activity)** and **Task 13**. It demonstrates the iterative NTLM credential failures (ATTEMPT=1 through ATTEMPT=4) culminating in NTLMSSP.AUTH_OK and subsequent share mapping.

SMB 3.0.2 Dialect Selection





- **Context for Report:** Place your SMB dialect capture here for **Task 10 (Negotiated SMB Dialect)**, showing the packet details pane cleanly highlighting **SMB 3.0.2 (0x0302)**.

Tree Connect Response for \\FS01\HR

```
Frame 1704: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits)
Ethernet II, Src: MS-NLB-PhysServer-24_20 (02:18:00:00:00:20), Dst: MS-NLB-PhysServer-24_01:01 (02:18:00:00:01:01)
Internet Protocol Version 4, Src: 10.18.0.20, Dst: 10.18.0.101
Transmission Control Protocol, Src Port: 445, Dst Port: 55029, Seq: 232, Ack: 331, Len: 84
NetBIOS Session Service
SMB2 (Server Message Block Protocol version 2), Tree Connect Response, MessageId 10
  SMB2 Header
    ProtocolId: 0xfe534d42
    Header Length: 64
    Credit Charge: 0
    NT Status: STATUS_SUCCESS (0x00000000)
    Command: Tree Connect (3)
    Credits granted: 1
    Flags: 0x00000001, Response
    Chain Offset: 0x00000000
    Message ID: 10
    Reserved: 0x0000feff
    Tree Id: 0x00006d4f  \\FS01\HR
    Session Id: 0x0000db33861a48a3 Acct:<NULL> Domain:<NULL> Host:<NULL>
    Signature: 00000000000000000000000000000000
    [Response to: 1702]
    [Time from request: 12.000000 milliseconds]
  Tree Connect Response (0x03)
```

- **Context for Report:** Place image_e01b07.png here for **Task 9 (Shares)** and **Task 11**, highlighting the expanded SMB2 Header, **NT Status: STATUS_SUCCESS**, and the mounted share \\FS01\HR.

SMB2 Create Request for \\FS01\Finance

```
▾ SMB2 Header
    ProtocolId: 0xfe534d42
    Header Length: 64
    Credit Charge: 0
    Channel Sequence: 0
    Reserved: 0000
    Command: Create (5)
    Credits requested: 1
  ▸ Flags: 0x00000000
    Chain Offset: 0x00000000
    Message ID: 108
    Reserved: 0x0000feff
  ▸ Tree Id: 0x00002101  \\FS01\Finance
  ▸ Session Id: 0x0000a18e20181115 Acct:<NULL> Domain:<NULL> Host:<NULL>
    Signature: 00000000000000000000000000000000
    [Response in: 41104]
    [Time to response: 8.000000 milliseconds]
▾ Create Request (0x05)
  ▸ StructureSize: 0x0039
    Oplock: No oplock (0x00)
    Impersonation level: Impersonation (2)
    Create Flags: 0x0000000000000000
    Reserved: 0000000000000000
  ▸ Access Mask: 0x0012019f
  ▸ File Attributes: 0x00000000
  ▸ Share Access: 0x00000007, Read, Write, Delete
    Disposition: Open (if file exists open it, else fail) (1)
  ▸ Create Options: 0x00000020
  ▾ [File Id: 563ae150-877d-2aa0-787e-efa8446d57db]
      [File Id Hash: 0xca15415a]
      [Frame handle opened: 41102]
```

- **Context for Report:** Place image_d54847.png alongside the previous image to illustrate the transition from authentication to active share traversal (Create Request for \\FS01\Finance).

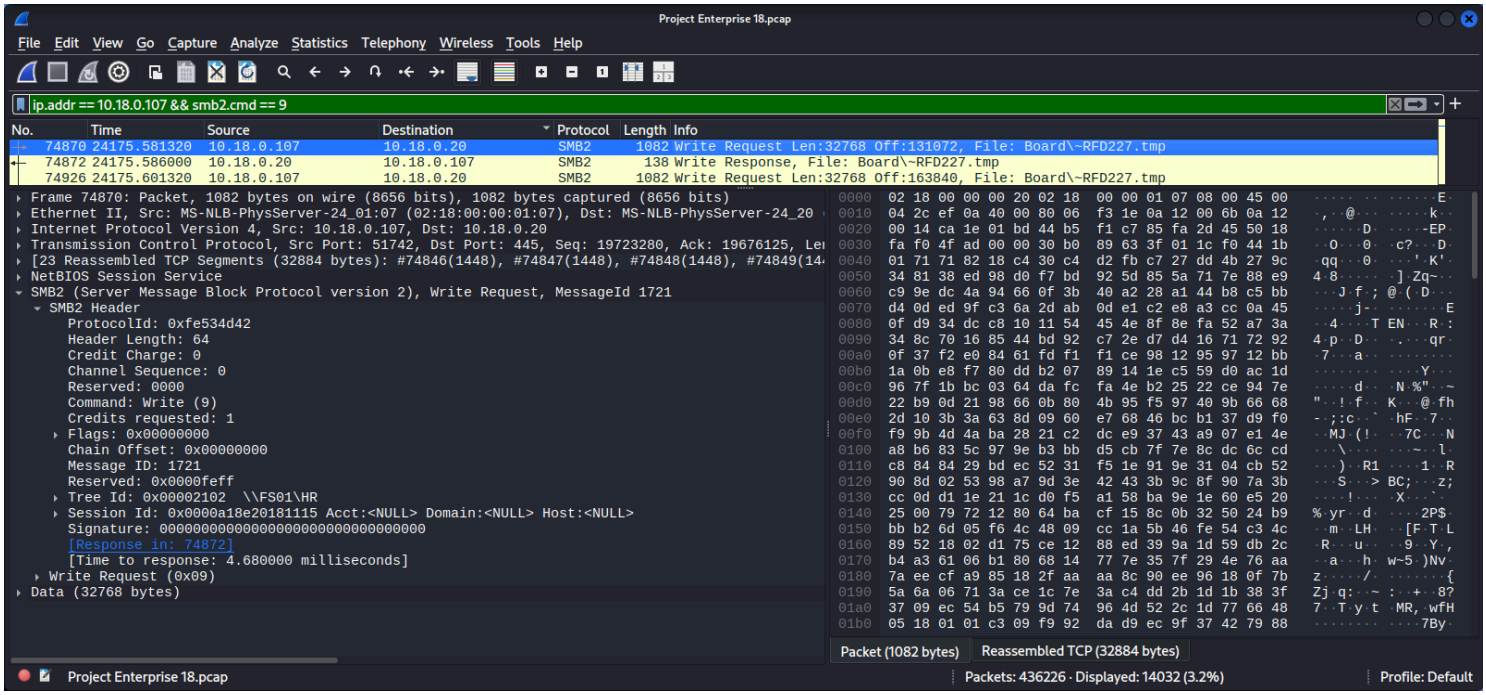
Access Denied Enforcement

```
▸ Frame 96288: Packet, 210 bytes on wire (1680 bits), 210 bytes captured (1680 bits)
▸ Ethernet II, Src: MS-NLB-PhysServer-24_20 (02:18:00:00:00:20), Dst: MS-NLB-PhysServer-24_01:07 (02:18:00:00:01:07)
▸ Internet Protocol Version 4, Src: 10.18.0.20, Dst: 10.18.0.107
▸ Transmission Control Protocol, Src Port: 445, Dst Port: 51742, Seq: 31588557, Ack: 31666029, Len: 156
▸ NetBIOS Session Service
▾ SMB2 (Server Message Block Protocol version 2), STATUS_ACCESS_DENIED, Create Response, MessageId 2750
  ▾ SMB2 Header
      ProtocolId: 0xfe534d42
      Header Length: 64
      Credit Charge: 0
      NT Status: STATUS_ACCESS_DENIED (0xc0000022)
      Command: Create (5)
      Credits granted: 1
    ▸ Flags: 0x00000001, Response
      Chain Offset: 0x00000000
      Message ID: 2750
      Reserved: 0x0000feff
    ▸ Tree Id: 0x00002102  \\FS01\HR
    ▸ Session Id: 0x0000a18e20181115 Acct:<NULL> Domain:<NULL> Host:<NULL>
      Signature: 00000000000000000000000000000000
      [Response to: 96286]
      [Time from request: 7.000000 milliseconds]
  ▸ Create Response (0x05)

NT Status code (smb2.nt_status), 4 bytes                                Pac
```

- **Context for Report:** Place image_e013dd.png under **Task 17**, proving server-side ACL enforcement via **NT Status: STATUS_ACCESS_DENIED (0xc0000022)** on restricted assets (Frame 96288).

High-Entropy Write Payload



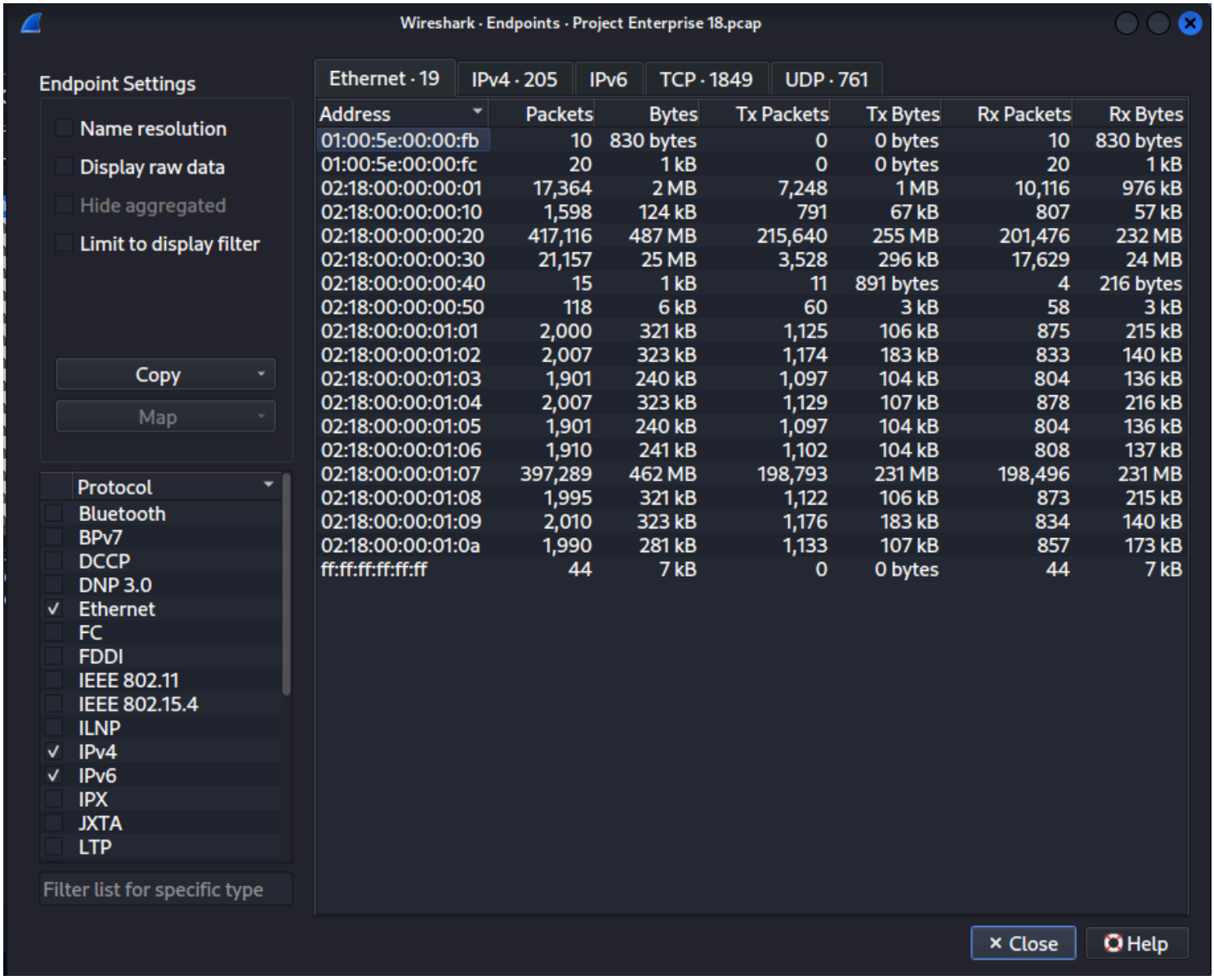
- **Context for Report:** Place image_d622dc.jpg under **Task 18**. Using filter ip.addr == 10.18.0.107 && smb2.cmd == 9, Frame 74870 displays staging to Board\~RFD227.tmp while the bottom-right ASCII pane reveals scrambled, high-entropy ciphertext.

4. Chronological Incident Timeline (10-Event Technical Breakdown)

1. $t = 24,059.01s$ — Initial SMB connection and Negotiate Protocol Request from 10.18.0.107 to 10.18.0.20.
2. $t = 24,059.05s$ — NTLM authentication probing resulting in STATUS_LOGON_FAILURE.
3. $t = 24,059.20s$ — Successful credential acceptance and multi-share tree connection (\\FS01\Finance, \\FS01\HR, \\FS01\Projects, \\FS01\Legal, \\FS01\Departments).
4. $t = 24,059.35s$ — Directory enumeration (Find Request with wildcard *) to catalog target files.
5. $t = 24,067.17s$ — Creation of temporary staging files (~RF*.tmp).
6. $t = 24,067.20s$ — Sequential READ commands ingesting 32 KB blocks into memory.
7. $t = 24,067.36s$ — High-volume WRITE commands replacing structured data with encrypted ciphertext.
8. $t = 24,067.40s$ — SET_INFO rename operations (SMB2_FILE_RENAME_INFO) appending the .arc2018 extension.
9. $t = 24,244.86s$ — Unauthorized access attempt on restricted assets yielding STATUS_ACCESS_DENIED.
10. $t = 25,368.80s$ — Conclusion of primary encryption loop and return to idle keep-alive traffic.

PART 3: Host-and-Service Map

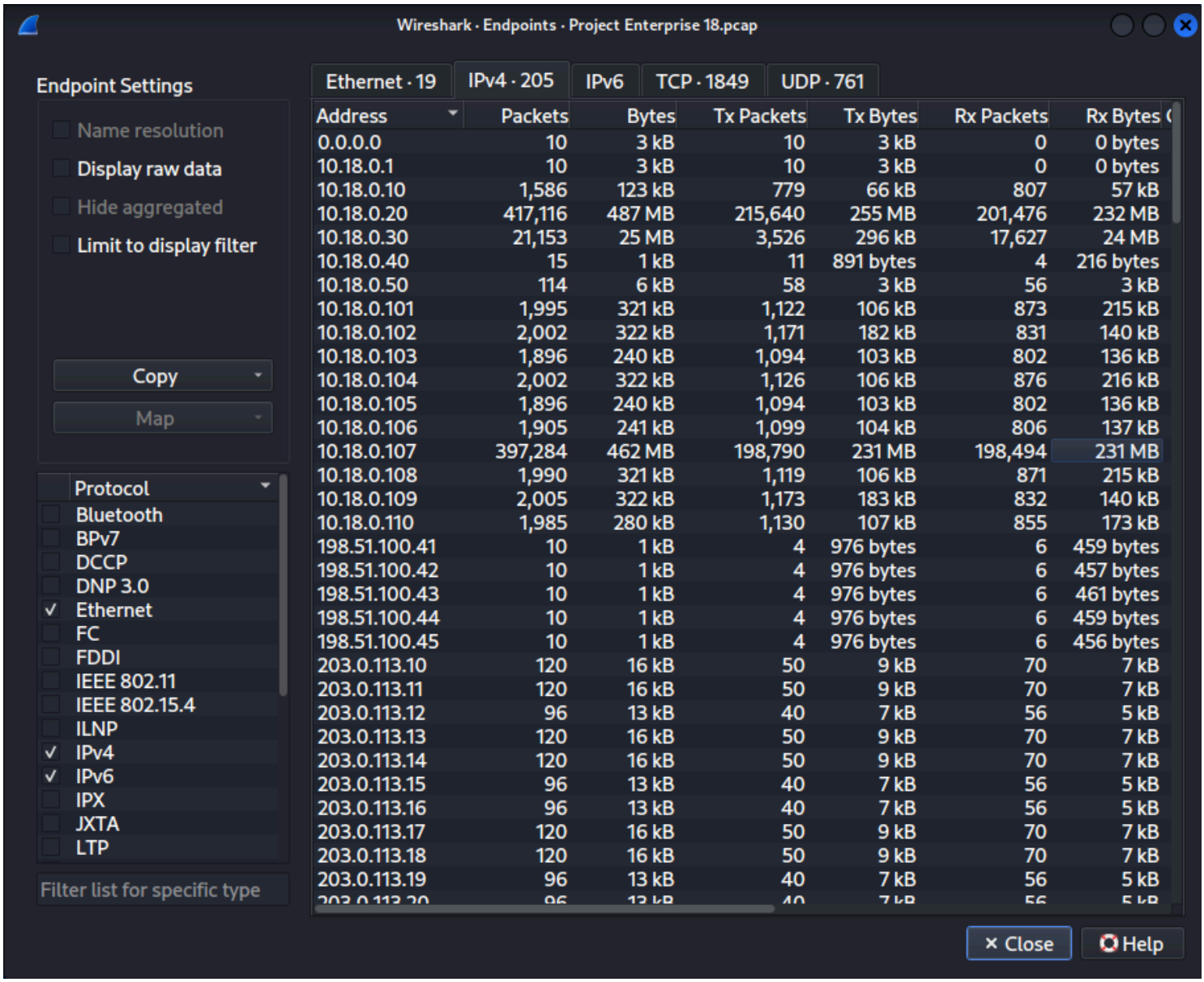
- **Internal Subnet:** 10.18.0.0/24
- **Primary File Server (FS01):** 10.18.0.20 (Moved 487 MB / 417,116 packets; provides SMB TCP/445 and NetBIOS TCP/139).
- **Compromised Workstation:** 10.18.0.107 (Moved 462 MB / 397,284 packets; executed automated encryption-in-place).
- **Legitimate Baseline Host:** 10.18.0.30 (Moved 25 MB / 21,143 packets between $t = 14,999s$ and $15,724s$; exhibited read-only sequential access matching routine backup profiles).
- **External Test Ranges:** 198.51.100.x & 203.0.113.x (RFC 5737 test destinations for background noise/telemetry).



PART 4: Confirmed Indicators, Rejected Leads & Unresolved Questions

Confirmed Indicators (IoCs)

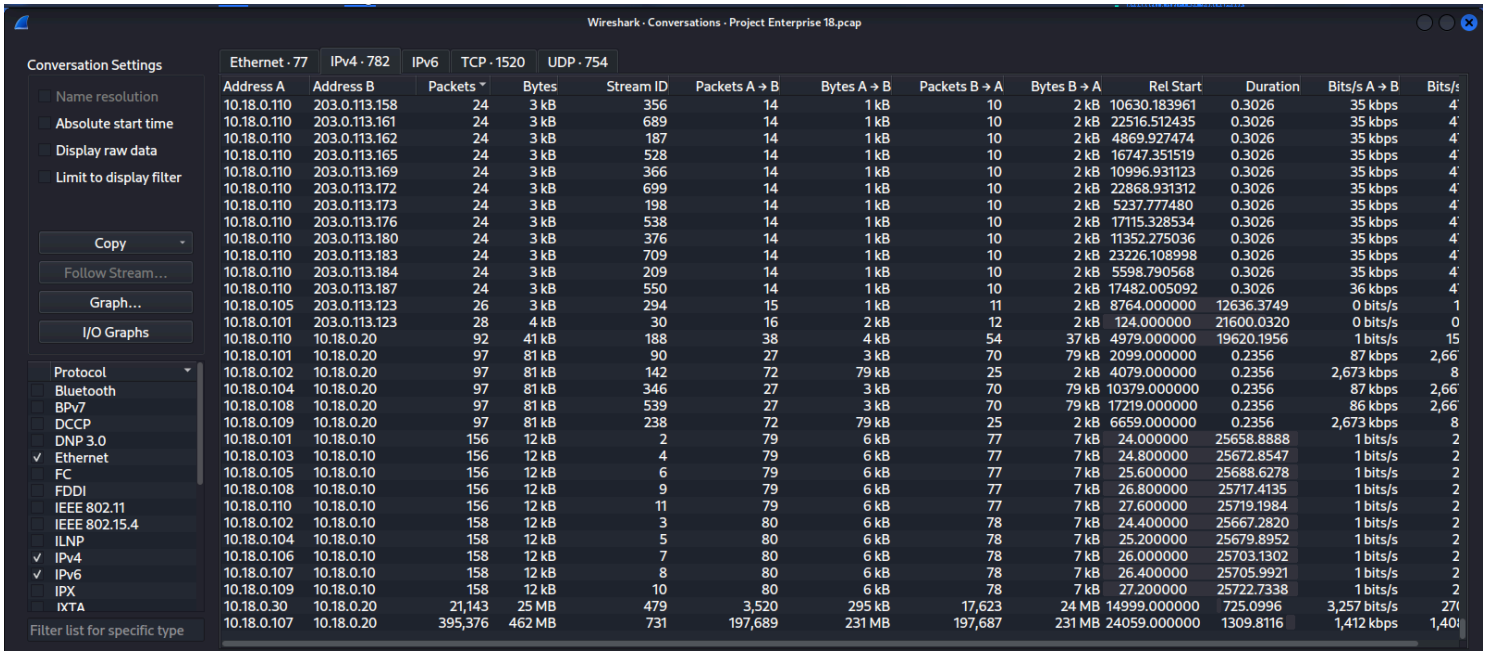
- **Attacker IP:** 10.18.0.107 | **Target Server IP:** 10.18.0.20 (FS01)
- **Target Shares:** \\FS01\Finance, \\FS01\HR, \\FS01\Projects, \\FS01\Legal, \\FS01\Departments
- **Staging File Format:** ~RF*.tmp (Temporary buffers)
- **Encrypted File Extension:** .arc2018
- **SMB Dialect:** SMB 3.0.2



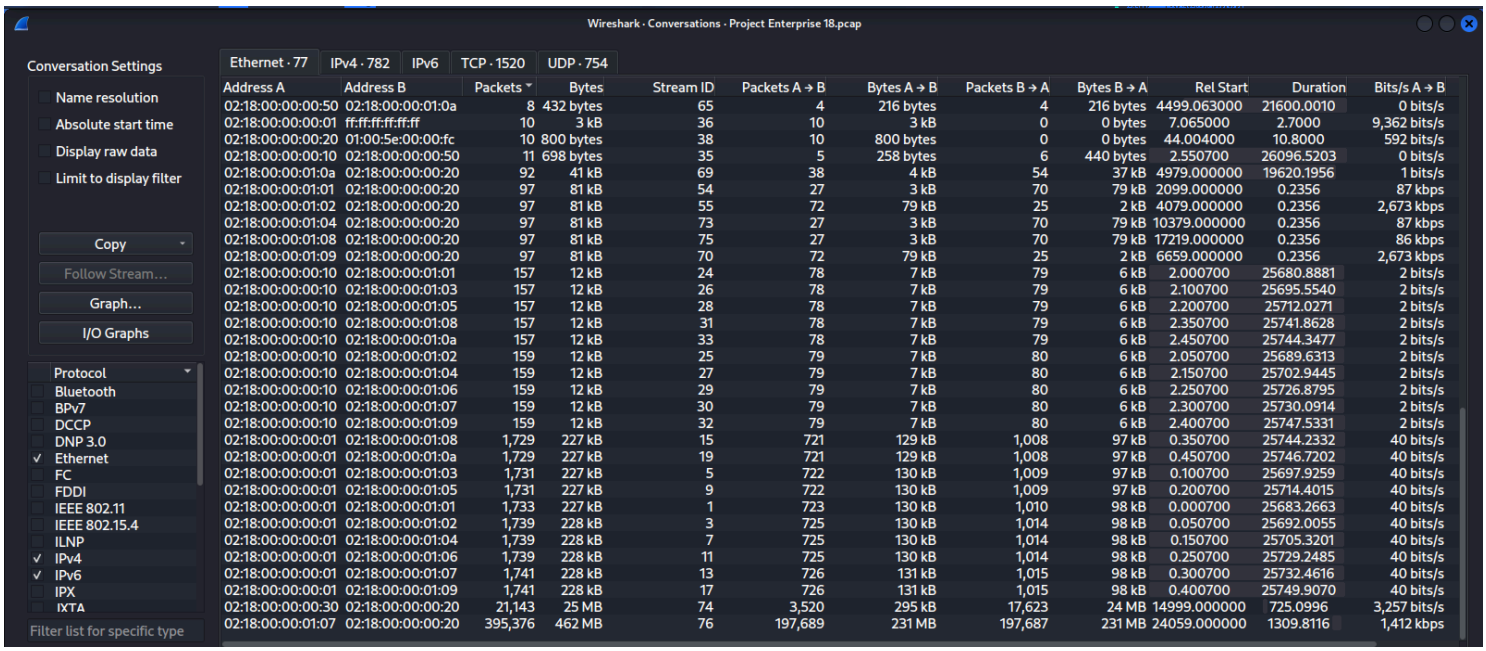
Rejected Leads (Alternative Explanations & Noise)

- *External DNS Red Herrings:* Domains (sowcar.com, z9ls.com, etc.) resolving to 198.51.100.x were rejected as non-causal background noise due to temporal decoupling from the attack window.

- *Routine Backup Hypothesis (Tasks 19–21):* Client 10.18.0.30 demonstrated legitimate, read-heavy sequential archiving (absence of WRITE and SET_INFO renames), proving that suspect client 10.18.0.107's write-heavy churn was malicious encryption rather than routine maintenance.
- *Data Exfiltration Check (Task 22):* Outbound data exfiltration was rejected because packet volume and destination port tracking confirmed all high-volume traffic remained strictly internal between the client and file server (encryption-in-place, zero external data theft).



Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s
10.18.0.110	203.0.113.158	24	3 kB	356	14	1 kB	10	2 kB	10630.183961	0.3026	35 kbps	4
10.18.0.110	203.0.113.161	24	3 kB	689	14	1 kB	10	2 kB	22516.512435	0.3026	35 kbps	4
10.18.0.110	203.0.113.162	24	3 kB	187	14	1 kB	10	2 kB	4869.927474	0.3026	35 kbps	4
10.18.0.110	203.0.113.165	24	3 kB	528	14	1 kB	10	2 kB	16747.351519	0.3026	35 kbps	4
10.18.0.110	203.0.113.169	24	3 kB	366	14	1 kB	10	2 kB	10996.931123	0.3026	35 kbps	4
10.18.0.110	203.0.113.172	24	3 kB	699	14	1 kB	10	2 kB	22868.931312	0.3026	35 kbps	4
10.18.0.110	203.0.113.173	24	3 kB	198	14	1 kB	10	2 kB	5237.777480	0.3026	35 kbps	4
10.18.0.110	203.0.113.176	24	3 kB	538	14	1 kB	10	2 kB	17115.328534	0.3026	35 kbps	4
10.18.0.110	203.0.113.180	24	3 kB	376	14	1 kB	10	2 kB	11352.275036	0.3026	35 kbps	4
10.18.0.110	203.0.113.183	24	3 kB	709	14	1 kB	10	2 kB	23226.108998	0.3026	35 kbps	4
10.18.0.110	203.0.113.184	24	3 kB	209	14	1 kB	10	2 kB	5598.790568	0.3026	35 kbps	4
10.18.0.110	203.0.113.187	24	3 kB	550	14	1 kB	10	2 kB	17482.005092	0.3026	36 kbps	4
10.18.0.105	203.0.113.123	26	3 kB	294	15	1 kB	11	2 kB	8764.000000	12636.3749	0 bits/s	1
10.18.0.101	203.0.113.123	28	4 kB	30	16	2 kB	12	2 kB	124.000000	21600.0320	0 bits/s	0
10.18.0.110	10.18.0.20	92	41 kB	188	38	4 kB	54	37 kB	4979.000000	19620.1956	1 bits/s	15
10.18.0.101	10.18.0.20	97	81 kB	90	27	3 kB	70	79 kB	2099.000000	0.2356	87 kbps	2,66
10.18.0.102	10.18.0.20	97	81 kB	142	72	79 kB	25	2 kB	4079.000000	0.2356	2,673 kbps	8
10.18.0.104	10.18.0.20	97	81 kB	346	27	3 kB	70	79 kB	10379.000000	0.2356	87 kbps	2,66
10.18.0.108	10.18.0.20	97	81 kB	539	27	3 kB	70	79 kB	17219.000000	0.2356	86 kbps	2,66
10.18.0.109	10.18.0.20	97	81 kB	238	72	79 kB	25	2 kB	6659.000000	0.2356	2,673 kbps	8
10.18.0.101	10.18.0.10	156	12 kB	2	79	6 kB	77	7 kB	24.000000	25658.8888	1 bits/s	2
10.18.0.103	10.18.0.10	156	12 kB	4	79	6 kB	77	7 kB	24.800000	25672.8547	1 bits/s	2
10.18.0.105	10.18.0.10	156	12 kB	6	79	6 kB	77	7 kB	25.600000	25688.6278	1 bits/s	2
10.18.0.108	10.18.0.10	156	12 kB	9	79	6 kB	77	7 kB	26.800000	25717.4135	1 bits/s	2
10.18.0.110	10.18.0.10	156	12 kB	11	79	6 kB	77	7 kB	27.600000	25719.1984	1 bits/s	2
10.18.0.102	10.18.0.10	158	12 kB	3	80	6 kB	78	7 kB	24.400000	25667.2820	1 bits/s	2
10.18.0.104	10.18.0.10	158	12 kB	5	80	6 kB	78	7 kB	25.200000	25679.8952	1 bits/s	2
10.18.0.106	10.18.0.10	158	12 kB	7	80	6 kB	78	7 kB	26.000000	25703.1302	1 bits/s	2
10.18.0.107	10.18.0.10	158	12 kB	8	80	6 kB	78	7 kB	26.400000	25705.9921	1 bits/s	2
10.18.0.109	10.18.0.10	158	12 kB	10	80	6 kB	78	7 kB	27.200000	25722.7338	1 bits/s	2
10.18.0.30	10.18.0.20	21,143	25 MB	479	3,520	295 kB	17,623	24 MB	14999.000000	725.0996	3,257 bits/s	27
10.18.0.107	10.18.0.20	395,376	462 MB	731	197,689	231 MB	197,687	231 MB	24059.000000	1309.8116	1,412 kbps	1,40



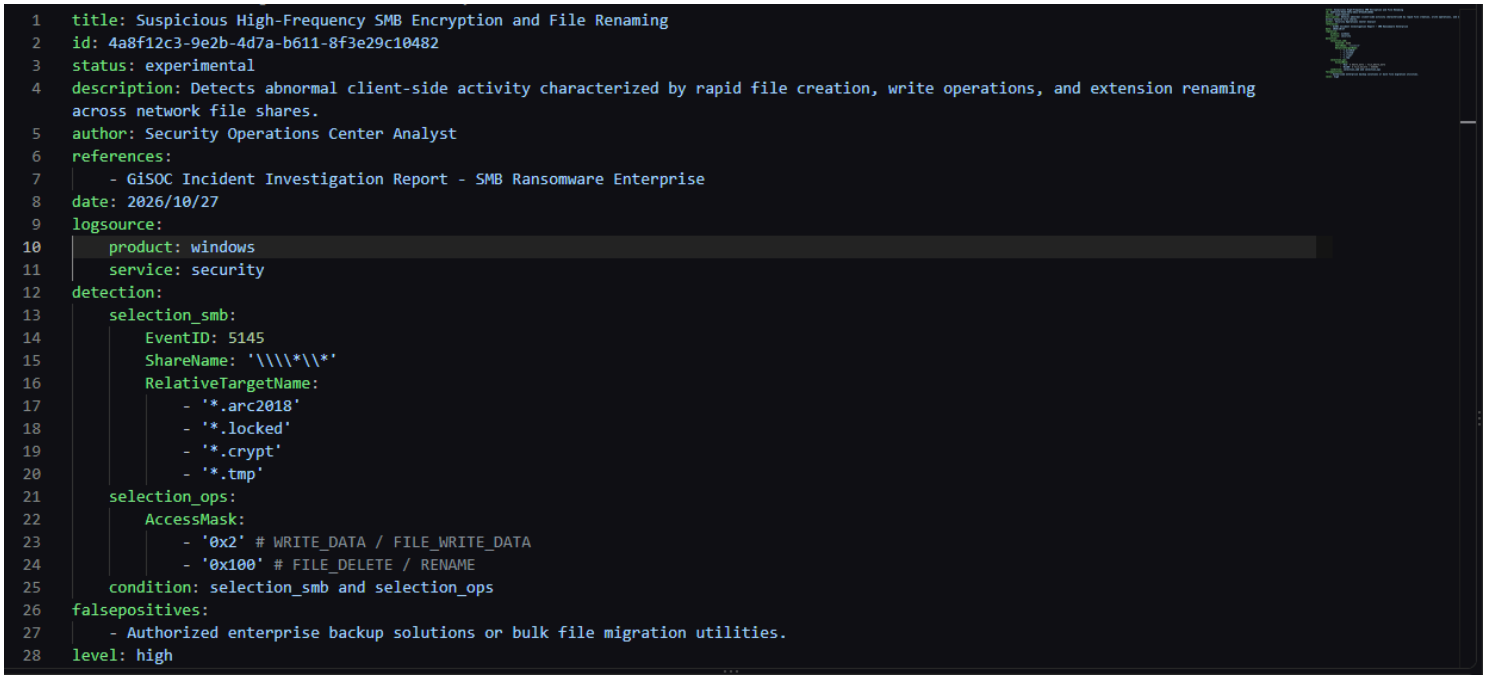
Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B
02:18:00:00:00:50	02:18:00:00:01:0a	8	432 bytes	65	4	216 bytes	4	216 bytes	4499.063000	21600.0010	0 bits/s
02:18:00:00:00:01	ff:ff:ff:ff:ff:ff	10	3 kB	36	10	3 kB	0	0 bytes	7.065000	2.7000	9,362 bits/s
02:18:00:00:00:20	01:00:5e:00:00:fc	10	800 bytes	38	10	800 bytes	0	0 bytes	44.004000	10.8000	592 bits/s
02:18:00:00:00:10	02:18:00:00:00:50	11	698 bytes	35	5	258 bytes	6	440 bytes	2.550700	26096.5203	0 bits/s
02:18:00:00:01:0a	02:18:00:00:00:20	92	41 kB	69	38	4 kB	54	37 kB	4979.000000	19620.1956	1 bits/s
02:18:00:00:01:01	02:18:00:00:00:20	97	81 kB	54	27	3 kB	70	79 kB	2099.000000	0.2356	87 kbps
02:18:00:00:01:02	02:18:00:00:00:20	97	81 kB	55	72	79 kB	25	2 kB	4079.000000	0.2356	2,673 kbps
02:18:00:00:01:04	02:18:00:00:00:20	97	81 kB	73	27	3 kB	70	79 kB	10379.000000	0.2356	87 kbps
02:18:00:00:01:08	02:18:00:00:00:20	97	81 kB	75	27	3 kB	70	79 kB	17219.000000	0.2356	86 kbps
02:18:00:00:01:09	02:18:00:00:00:20	97	81 kB	70	72	79 kB	25	2 kB	6659.000000	0.2356	2,673 kbps
02:18:00:00:00:10	02:18:00:00:01:01	157	12 kB	24	78	7 kB	79	6 kB	2.000700	25680.8881	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:03	157	12 kB	26	78	7 kB	79	6 kB	2.100700	25695.5540	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:05	157	12 kB	28	78	7 kB	79	6 kB	2.200700	25712.0271	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:08	157	12 kB	31	78	7 kB	79	6 kB	2.350700	25741.8628	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:0a	157	12 kB	33	78	7 kB	79	6 kB	2.450700	25744.3477	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:02	159	12 kB	25	79	7 kB	80	6 kB	2.050700	25689.6313	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:04	159	12 kB	27	79	7 kB	80	6 kB	2.150700	25702.9445	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:06	159	12 kB	29	79	7 kB	80	6 kB	2.250700	25726.8795	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:07	159	12 kB	30	79	7 kB	80	6 kB	2.300700	25730.0914	2 bits/s
02:18:00:00:00:10	02:18:00:00:01:09	159	12 kB	32	79	7 kB	80	6 kB	2.400700	25747.5331	2 bits/s
02:18:00:00:00:01	02:18:00:00:01:08	1,729	227 kB	15	721	129 kB	1,008	97 kB	0.350700	25744.2332	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:0a	1,729	227 kB	19	721	129 kB	1,008	97 kB	0.450700	25746.7202	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:03	1,731	227 kB	5	722	130 kB	1,009	97 kB	0.100700	25697.9259	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:05	1,731	227 kB	9	722	130 kB	1,009	97 kB	0.200700	25714.4015	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:01	1,733	227 kB	1	723	130 kB	1,010	98 kB	0.000700	25683.2663	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:02	1,739	228 kB	3	725	130 kB	1,014	98 kB	0.050700	25692.0055	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:04	1,739	228 kB	7	725	130 kB	1,014	98 kB	0.150700	25705.3201	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:06	1,739	228 kB	11	725	130 kB	1,014	98 kB	0.250700	25729.2485	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:07	1,741	228 kB	13	726	131 kB	1,015	98 kB	0.300700	25732.4616	40 bits/s
02:18:00:00:00:01	02:18:00:00:01:09	1,741	228 kB	17	726	131 kB	1,015	98 kB	0.400700	25749.9070	40 bits/s
02:18:00:00:00:30	02:18:00:00:00:20	21,143	25 MB	74	3,520	295 kB	17,623	24 MB	14999.000000	725.0996	3,257 bits/s
02:18:00:00:01:07	02:18:00:00:00:20	395,376	462 MB	76	197,689	231 MB	197,687	231 MB	24059.000000	1309.8116	1,412 kbps

Unresolved Questions

- What initial access vector (e.g., phishing lure, drive-by download) triggered the payload on workstation 10.18.0.107?
- How were valid enterprise credentials acquired prior to the initial NTLM session establishment?

PART 5: Proposed Behavioral Detection Rule / Analytic

Below is a screenshot of the code on my Vs Code Terminal



```yaml
title: Suspicious High-Frequency SMB Encryption and File Renaming
id: 4a8f12c3-9e2b-4d7a-b611-8f3e29c10482
status: experimental
description: Detects abnormal client-side activity characterized by rapid file creation, write operations, and extension renaming
across network file shares.
author: Security Operations Center Analyst
references:
    - GiSOC Incident Investigation Report - SMB Ransomware Enterprise
date: 2026/10/27
logsource:
    product: windows
    service: security
detection:
    selection_smb:
        EventID: 5145
        ShareName: '\\\\*\\*'
        RelativeTargetName:
            - '*.arc2018'
            - '*.locked'
            - '*.crypt'
            - '*.tmp'
    selection_ops:
        AccessMask:
            - '0x2' # WRITE_DATA / FILE_WRITE_DATA
            - '0x100' # FILE_DELETE / RENAME
    condition: selection_smb and selection_ops
falsepositives:
    - Authorized enterprise backup solutions or bulk file migration utilities.
level: high
```